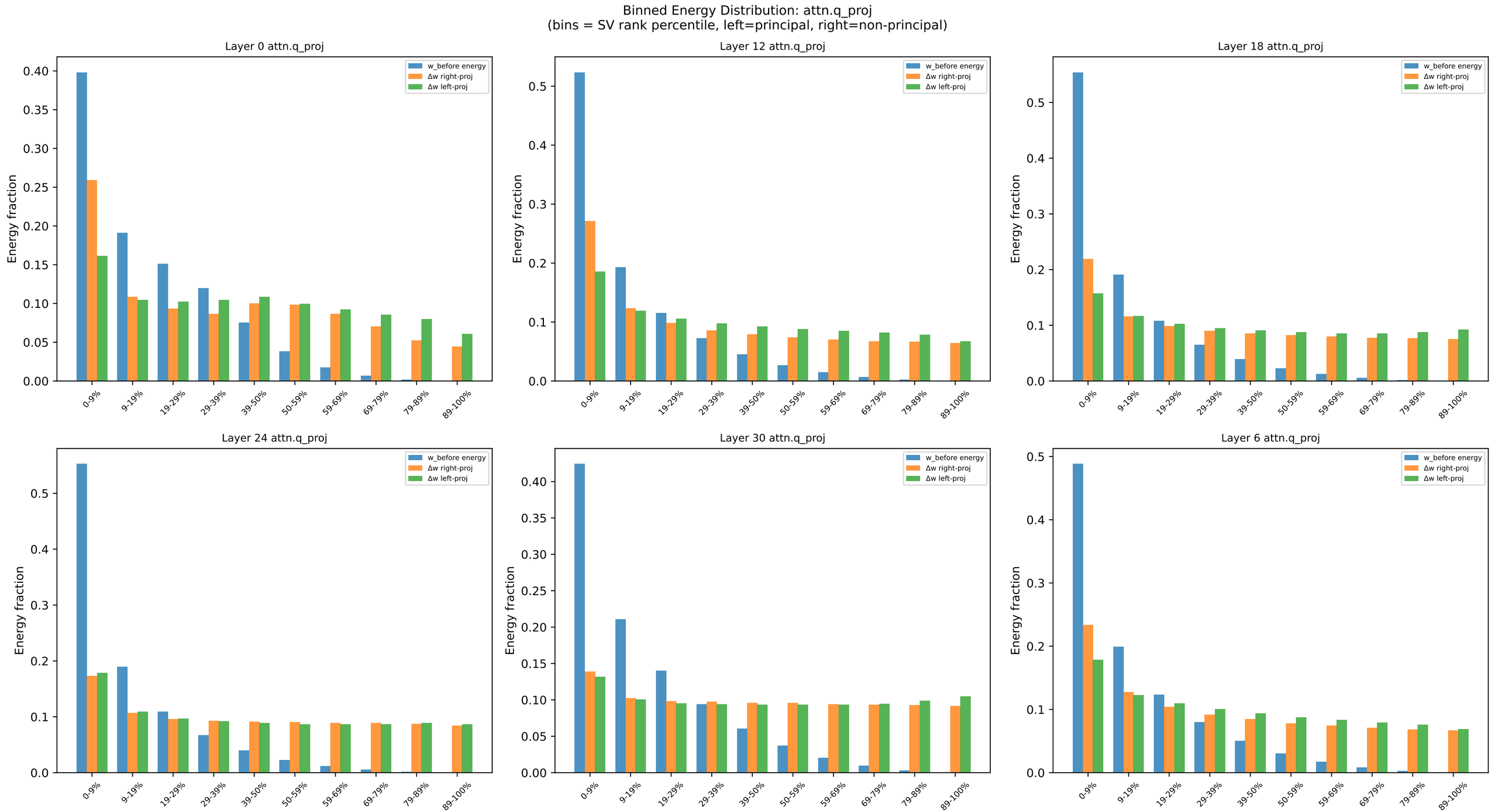
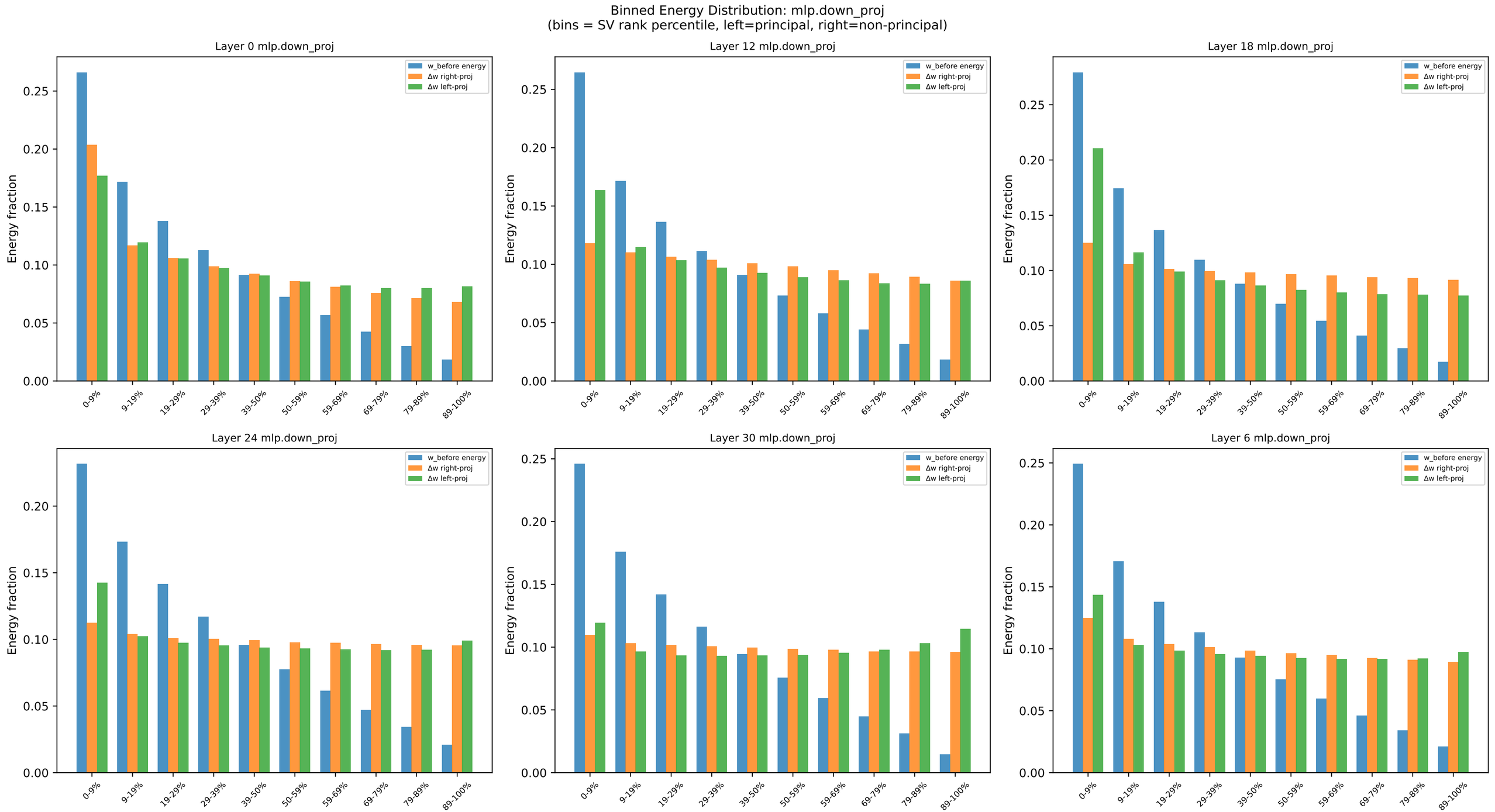
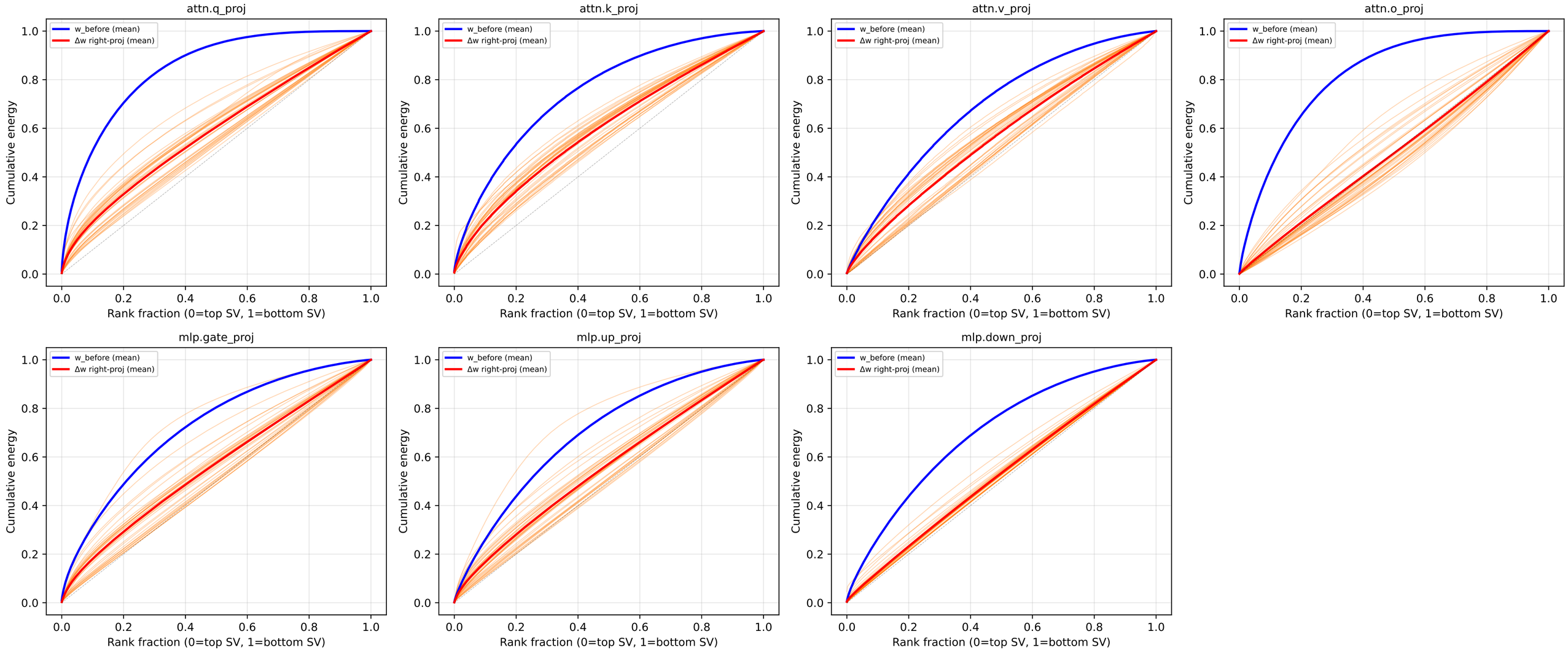
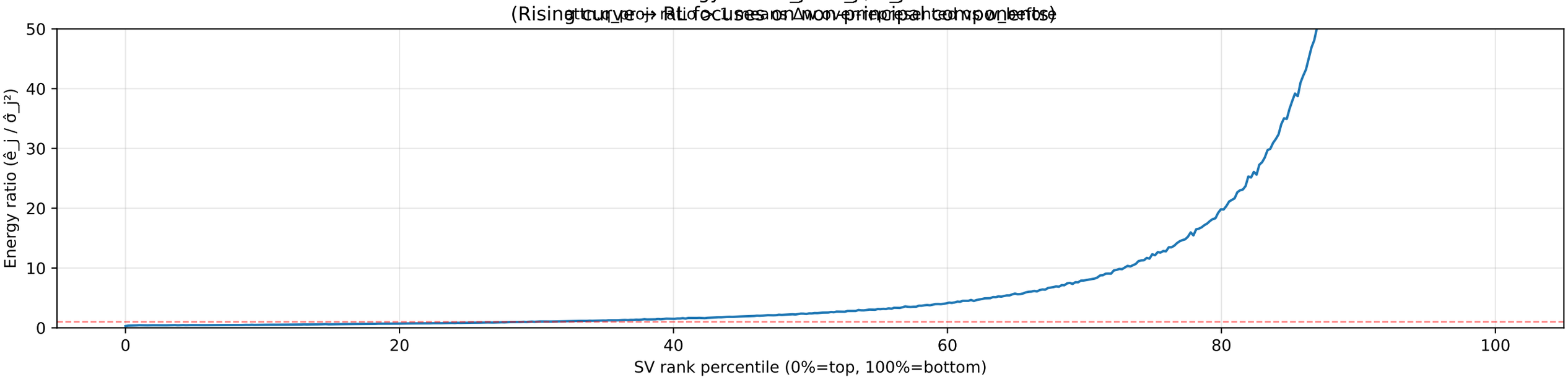
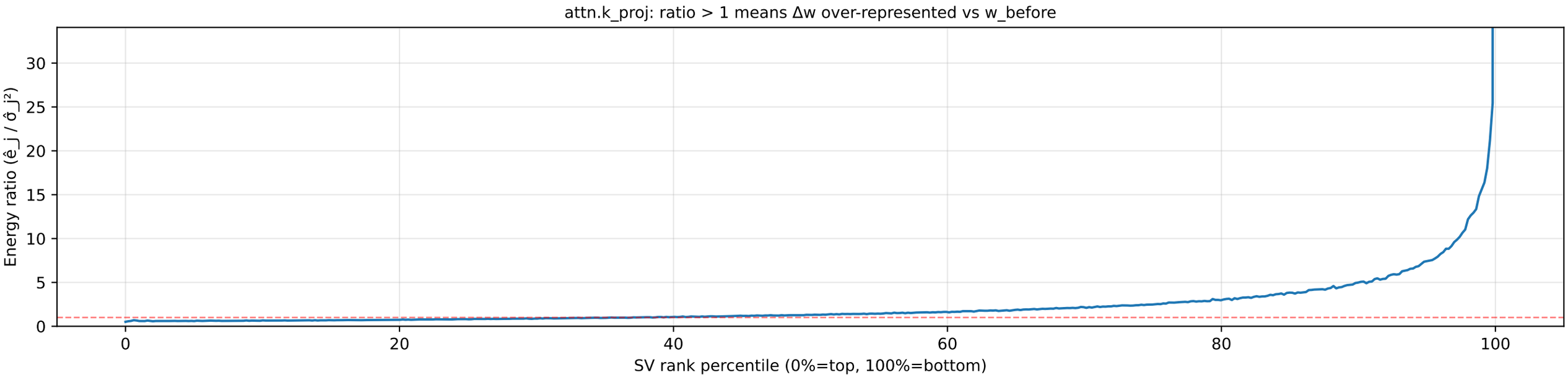

Binned Energy Distribution: attn.q_proj
(bins = SV rank percentile, left=principal, right=non-principal)
Layer 0 attn.q_proj
Layer 12 attn.q_proj
Layer 18 attn.q_proj
Layer 24 attn.q_proj
Layer 30 attn.q_proj
Layer 6 attn.q_proj
w_before energy
Δw right-proj
Δw left-proj
Energy fraction
0-9%
9-19%
19-29%
29-39%
39-50%
50-59%
59-69%
69-79%
79-89%
89-100%

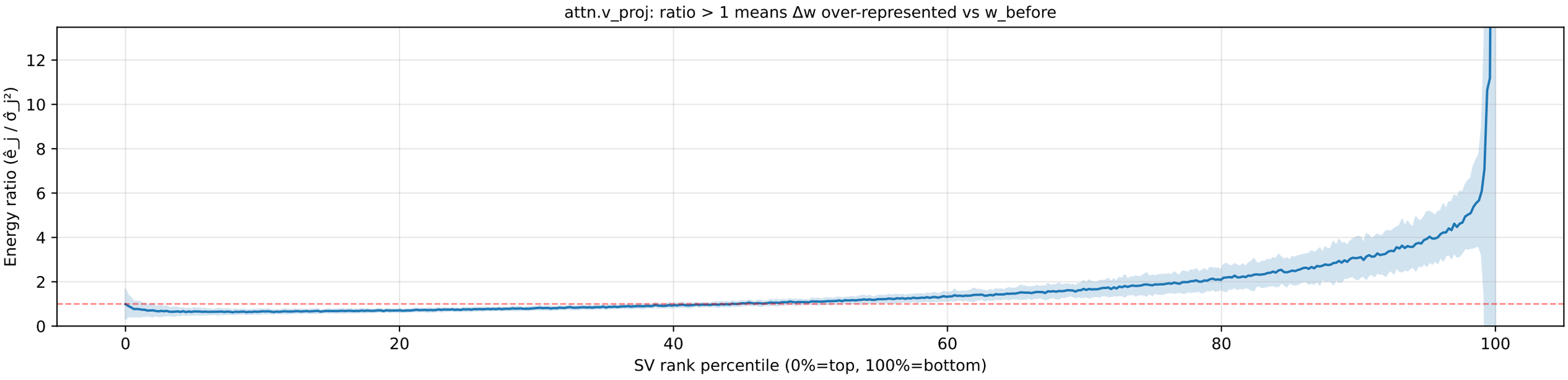

Binned Energy Distribution: mlp.down_proj
(bins = SV rank percentile, left=principal, right=non-principal)
Layer 0 mlp.down_proj
Layer 12 mlp.down_proj
Layer 18 mlp.down_proj
Layer 24 mlp.down_proj
Layer 30 mlp.down_proj
Layer 6 mlp.down_proj
w_before energy
Δw right-proj
Δw left-proj
Energy fraction
0.00
0.05
0.10
0.15
0.20
0.25
0-9%
9-19%
19-29%
29-39%
39-50%
50-59%
59-69%
69-79%
79-89%
89-100%

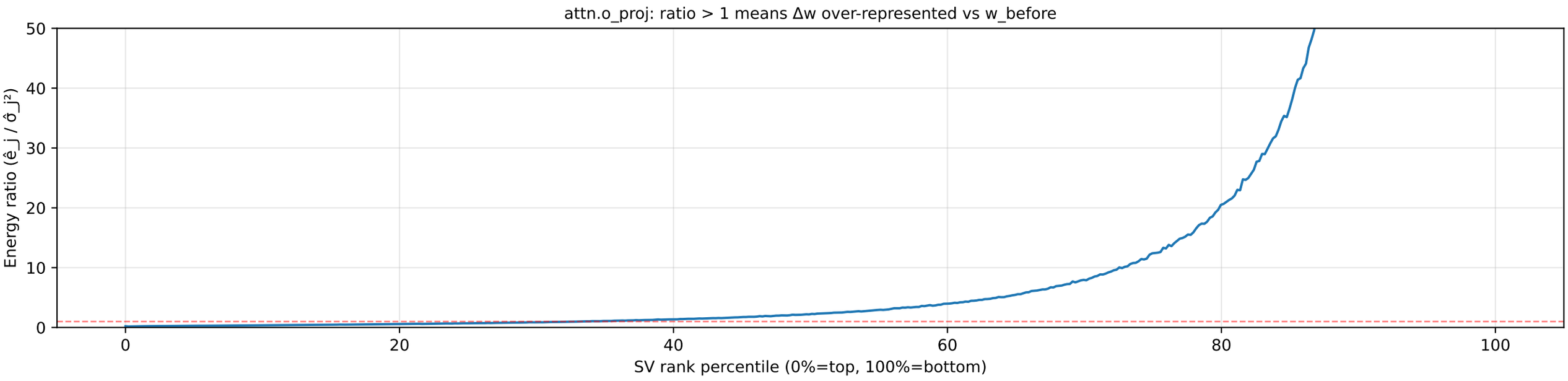

Cumulative Energy: w_before vs Δw projection
(If red curve is below blue → Δw concentrates on non-principal components)
attn.q_proj
attn.k_proj
attn.v_proj
attn.o_proj
mlp.gate_proj
mlp.up_proj
mlp.down_proj
w_before (mean)
Δw right-proj (mean)
Cumulative energy
Rank fraction (0=top SV, 1=bottom SV)
0.0
0.2
0.4
0.6
0.8
1.0

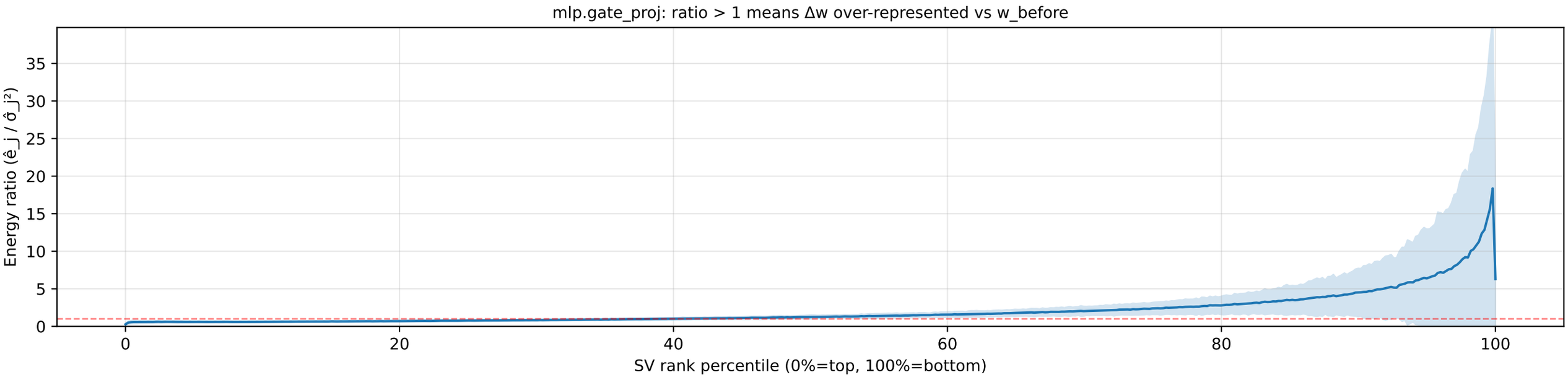
Energy Ratio r_j = ê_j / σ̂_j²
(Rising curve → Ratio > 1 means Δw over-represented in principal components)

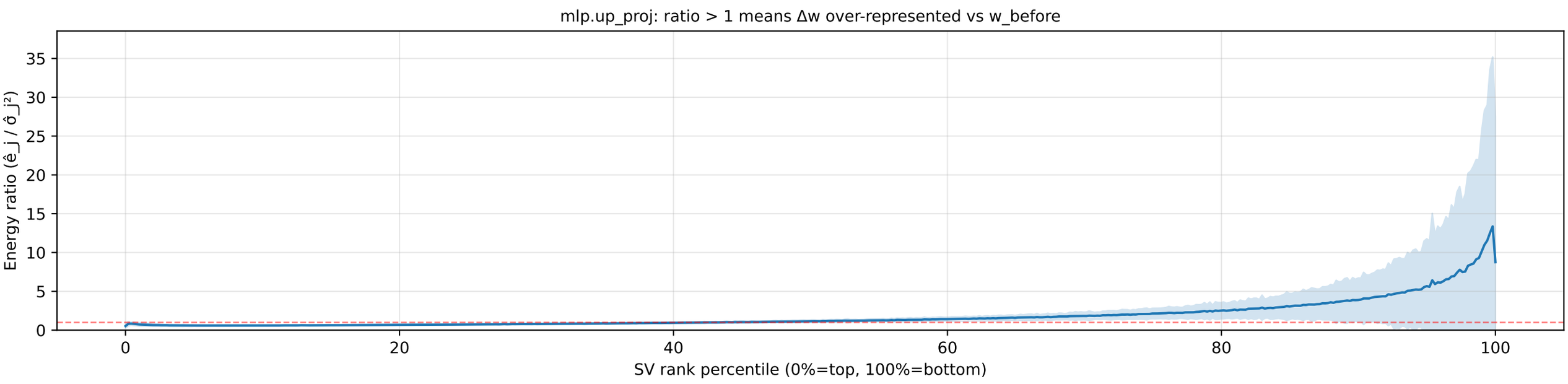


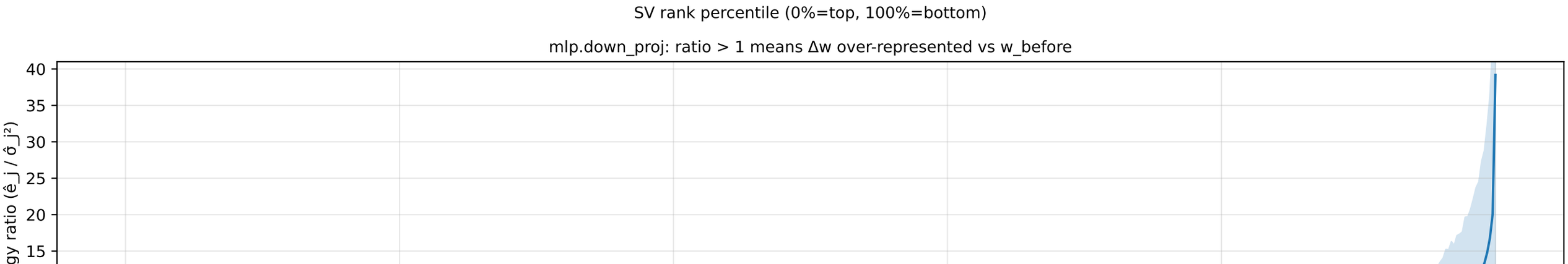



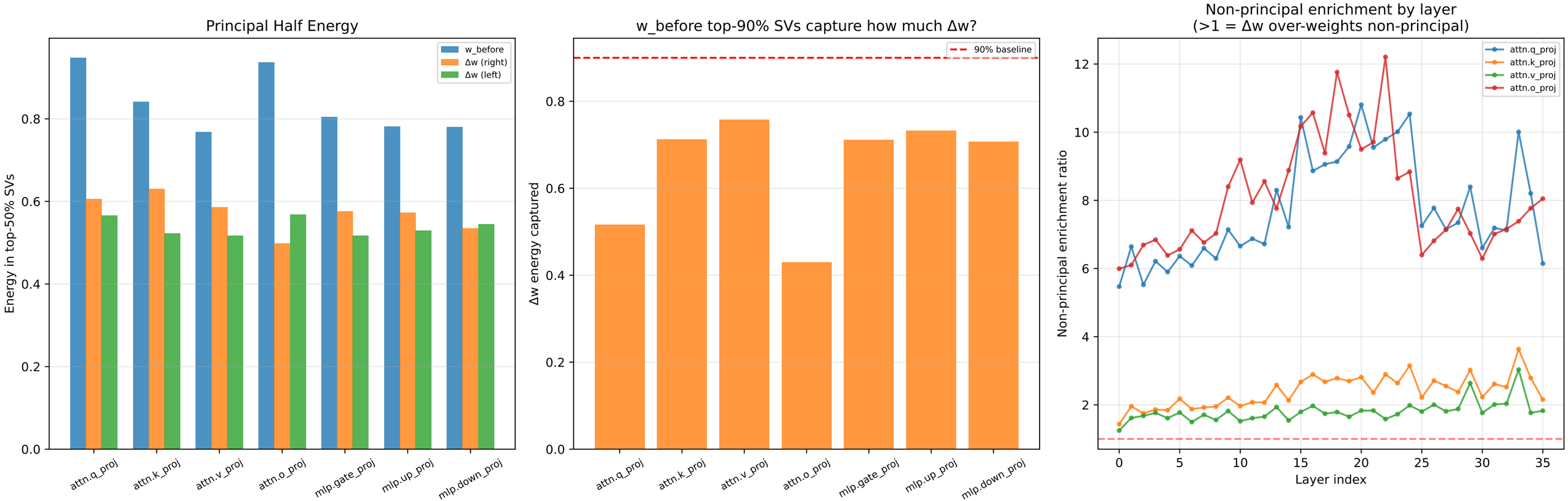



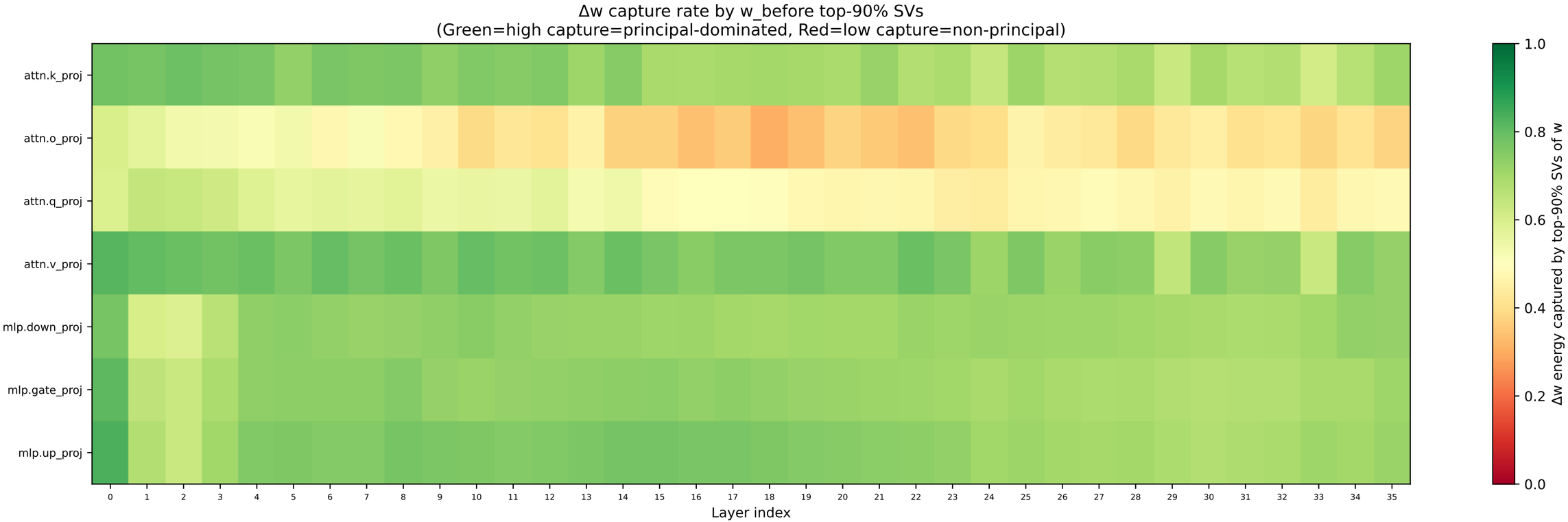



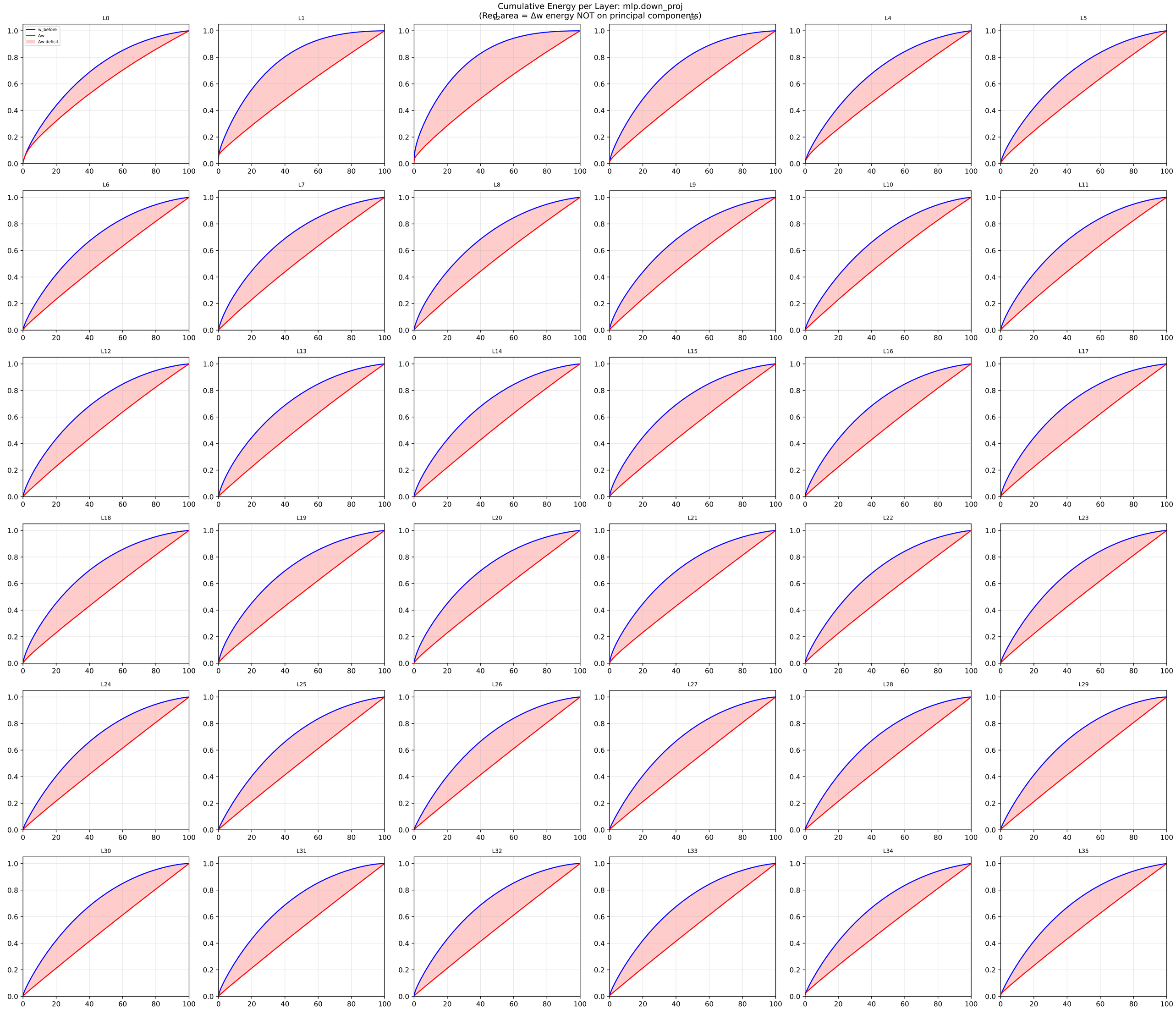




Principal Half Energy
Energy in top-50% SVs
w_before
Δw (right)
Δw (left)
0.0
0.2
0.4
0.6
0.8
attn.q_proj
attn.k_proj
attn.v_proj
attn.o_proj
mlp.gate_proj
mlp.up_proj
mlp.down_proj
w_before top-90% SVs capture how much Δw?
Δw energy captured
90% baseline
Non-principal enrichment by layer
(>1 = Δw over-weights non-principal)
Non-principal enrichment ratio
Layer index
attn.q_proj
attn.k_proj
attn.v_proj
attn.o_proj
2
4
6
8
10
12
0
5
10
15
20
25
30
35


Δw capture rate by w_before top-90% SVs
(Green=high capture=principal-dominated, Red=low capture=non-principal)
attn.k_proj
attn.o_proj
attn.q_proj
attn.v_proj
mlp.down_proj
mlp.gate_proj
mlp.up_proj
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
Layer index
1.0
0.8
0.6
0.4
0.2
0.0
Δw energy captured by top-90% SVs of w


Cumulative Energy per Layer: mlp.down_proj
(Red area = Δw energy NOT on principal components)
w_before
Δw
Δw deficit
L0
L1
L2
L3
L4
L5
L6
L7
L8
L9
L10
L11
L12
L13
L14
L15
L16
L17
L18
L19
L20
L21
L22
L23
L24
L25
L26
L27
L28
L29
L30
L31
L32
L33
L34
L35